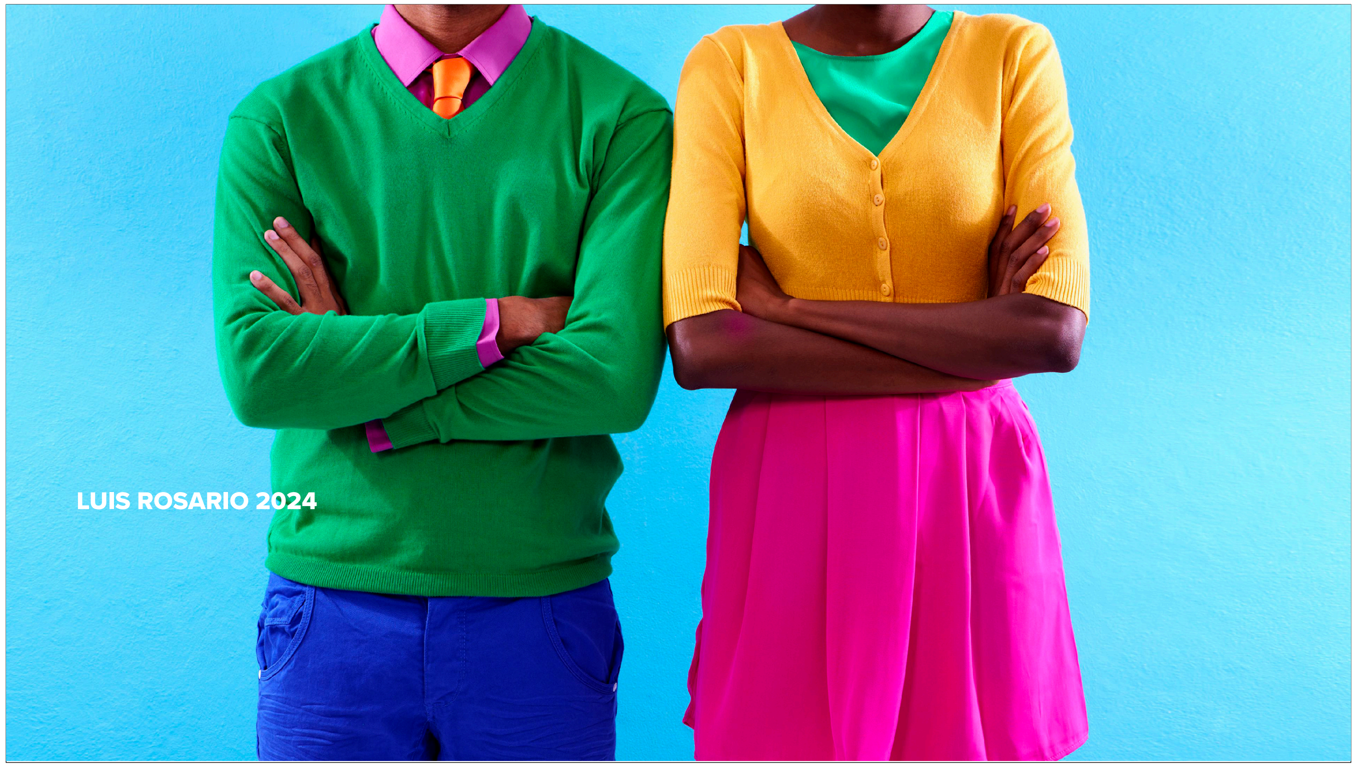
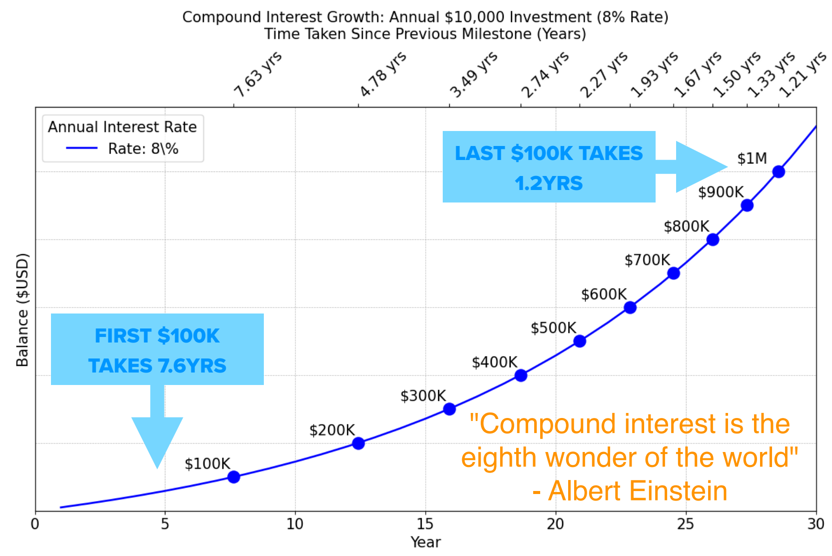

LUIS ROSARIO 2024

DISCLAMER

- I am NOT endorsed by anyone and I am not promoting anyone.
- This is for educational and entertainment purposes only and NOT financial advice.
- I am an Engineer, not a CPA or CFA.

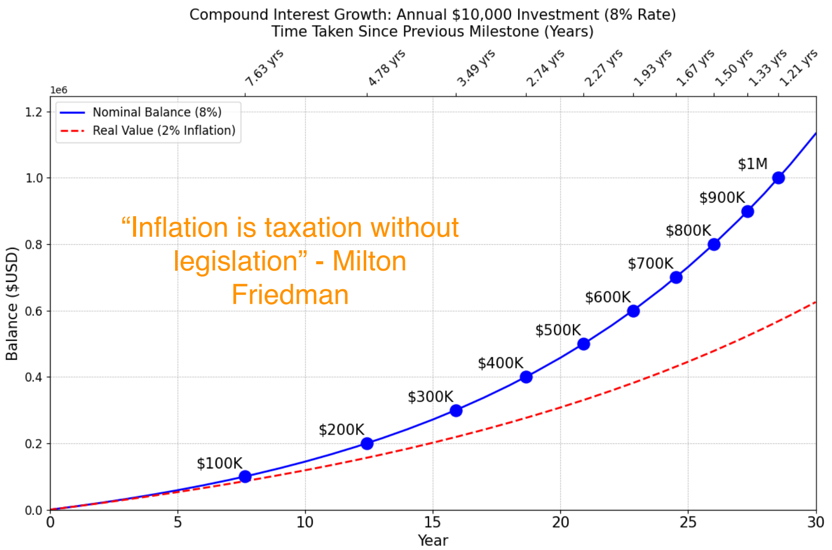
POWER OF COMPOUNDING INTEREST

- Most people earning $85,000 a year believe that their only shot at becoming a millionaire is to win the lottery. However, the power of compound interest illustrates how anyone can slowly transform investment in a large fortunes over time.
- How much money do you need to invest to reach $1M if you start investing when you are 10 years old and want to use the money when you are 65 years old? Answer



POWER OF INFLATION

- Money losses its purchasing power over time (aka is worth less)
- Saving money in the bank/under the mattress is not a good idea due to inflation
- Using the CPI, what cost \$1 in 1970 took \$6.19 to buy in 2014. A six-fold increase. In the same time period, a 4-year state school college education went from \$4,800 to \$160,000. A 33-fold increase.



https://www.calculator.net/inflation-calculator.html

INVESTMENT VEHICLES

- Work Retirement
  - 401k
  - TSP
  - 403b
  - 457
- Health Savings
  - Flexible Saving Account
  - Limited Flexible Saving Account
  - Health Saving Account
- Personal Retirement
  - IRA
    - Traditional
    - ROTH
  - Brokerage Accounts
- Crypto
- Options
- Annuities
- Life Insurance
- Stocks
- 529 Saving Plans
- Money Market Account
- Money Market Fund
- Certificates of Deposit
- Bonds
- TIPS
- I Bonds
- Cash Mgmt. Account

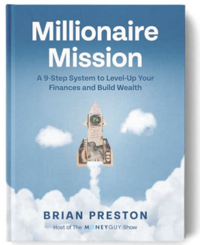
INVESTMENT ACCOUNTS SUMMARY

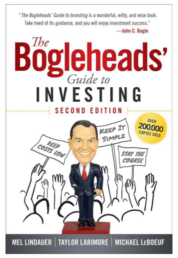
	Maximum (2026)***	Employer Contribution	RMD	Income Limit	Access Age	Taxes
Traditional 401k	$25000	Yes	Yes	No	59 1/2	Deferred
Roth 401k	$25000	No	No	No	59 1/2	2x Tax free
Traditional IRA	$7500	No	Yes	Yes	59 1/2	Deferred
Roth IRA	$7500	No	No	Yes	59 1/2	2x Tax Free
HSA	$4500*/$8550**	Yes	No	No	Retroactive	3x Tax Free
LP-FSA	$3400	No	No	No	Instant	2x Free
529	$19k*/$38k**	No	No	Yes	Instant	2x Free
Brokerage Acc.	Unlimited	No	No	No	Instant	Taxable
Cash Mgmt.	Unlimited	No	No	No	Instant	Taxable

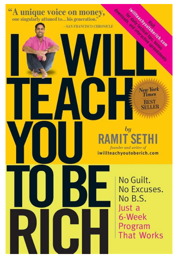
*For a single under 55, **Married under 55, ***Age and civil status will vary contributions

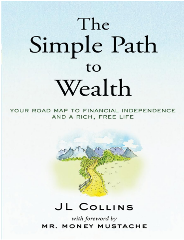
HOW MUCH TO INVEST? SAVING...

- Streamline Expenses
  - Improve Credit Score
  - Refinance Car, Mortgage with lower rates
  - Debt Consolidation
  - Pay Off Debt
  - Don't carry a credit card balance
  - Shop for insurance
- Goals, Health and Age
  - Make up period
  - Accumulation/Preservation







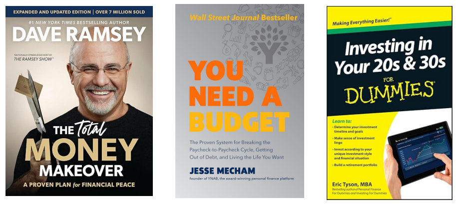


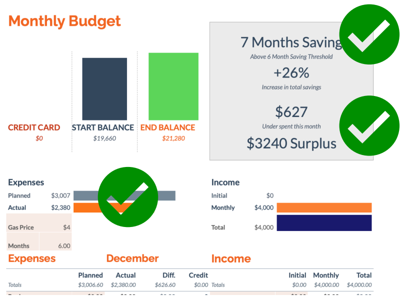
"If you intend to achieve financial freedom, you are going to have to think differently. It starts by recognizing that debt should not be considered normal. It should be recognized as the vicious, pernicious destroyer of wealth-building potential it truly is. It has no place in your financial life... the more debt you carry the more of your income is devoured by interest payments." - JL Collin

IT IS NOT HOW MUCH YOU MAKE BUT HOW MUCH YOU CAN SAVE.

HOW MUCH TO INVEST? BUDGETING...

- Budget Apps
- Track Expenses (i.e. subscriptions, cell-phones)
- Zero-Based Budgeting and cash flow
- Avoid Lifestyle Creep
  - Downsize house or car and memberships
  - Use a library card for audio books and movies
- Tax efficiency and tax planning
- Avoid fees (late, bank, credit card, utilities, annual fees) using Auto-payment
- All in one app (Saving Planner, Cash Flow, Budgeting, etc), WPCU, EMPOWER, Chase, etc...





"SUPPOSE ONE OF YOU WANTS TO BUILD A TOWER (*STABLE FINANCIAL FUTURE*). WON'T YOU FIRST SIT DOWN AND ESTIMATE THE COST TO SEE IF YOU HAVE ENOUGH MONEY TO COMPLETE IT?"

HOW TO INVEST? ORDER OF INVESTMENT

- Both frameworks have their merits and place in investing. Depending of your mindset and financial situation you might need the Baby Steps and could transition to the FOO later.
- Watch some videos on comparing these two methods to get more familiar with the thought process of how to plan for the future
  - Video 1
  - Video 2
  - Video 3

Caution one size does not fit all. Low interest rate debt payment vs Investment

A GOOD MAN LEAVES AN INHERITANCE TO HIS CHILDREN'S CHILDREN… -PROVEEBS 13:22

1. The first step to Building Wealth isn't about chasing Investments or doubling your income it's about protecting yourself from financial disaster. A single unexpected event can derail everything. That's why the first mission is to keep your financial life out of the ditch by preparing for the inevitable emergencies that life throws your way. The focus here is on one essential principle, cover your highest deductible. Think of your deductible as the amount of money you need to pay out of pocket before before your insurance kicks in whether it's your health insurance, car insurance or homeowners policy.

2. "Never Leave free money on the table". If your employer offers a match on your retirement contributions maximizing it is one of the smartest financial decisions you can make. Its a guaranteed returns. For example if your company matches 100% of your contributions up to 5% of your salary that's an immediate 100% return on the money you put in. No investment in the stock market, real estate or elsewhere can guarantee returns like that. Your credit card is charging you 20%-30%, employer matching is offering 100% secure.
   See graph

3. One of the most powerful steps to achieving Financial Freedom is paying off high interest debt (8%+). Debt is one of the greatest barriers to Building Wealth and the longer you carry it the more it drains your financial future.

   Does my mortgage count as high-interest debt? Rates over 6% or even 7%. While the interest rate is bordering on what would be considered a high rate, there are a few reasons it may still count as low-interest debt. Mortgages are on homes, which are typically appreciating assets, a distinct difference from car loans, student loans, and consumer debt. Mortgage interest may be deductible if you itemize your tax deductions, which effectively lowers your interest rate.

4. This extends step one. Whether it's a medical emergency, car repair, job loss, a major home repair these situations are often out of our control and without an emergency Reserve they can quickly spiral into Financial chaos. Think beyond the typical advice of simply saving a few months of expenses. A true emergency Reserve is designed not only to handle minor bumps in the road but also to give you the flexibility and freedom to weather larger storms. The idea is to build a reserve that's large enough to protect your lifestyle without forcing you to dip into other saving or worse rack up high interest debt. When the unexpected happens how much do you need to stay calm? The rule of thumb here is to save at least 3 to 6 months of living expenses but depending on your personal situation, such as whether you're single, married or have dependence. You may need more if you have a less stable job or work in an industry prone to layoffs.

5. Max out tax-free growth with tax free accounts. The journey to wealth it's not just about how much you earn, it's about how much you keep. One of the most powerful ways to grow your wealth is by minimizing the taxes you pay from your Investments at retrieval. Shield your money from the tax man by maxing out tax-free growth opportunities through Roth IRA and health savings accounts (HSA) contributions. Taxes can be the silent destroyer of wealth as you build your financial portfolio taxes chip away at your earnings reducing the power of your Investments. The key to overcoming this challenge is understanding and utilizing tax-free growth accounts which allow your Investments to grow without the burden of taxes eating into your gains two of the most effective tools for this purpose are Roth IRAs and HSA.

6. Step 6 is in the same tax efficiency mindset. Employer sponsored retirements accounts allows you to take full advantage of tax deferred growth meaning your money compounds faster because it's not being eroded by taxes year after year. It also highlights how pre-tax contributions reduce your taxable income today giving you an immediate financial benefit while you save for the longterm. While traditional retirement accounts offer tax deferred growth, Roth accounts flip the equation your investments grow tax-free and withdrawals at retirement are completely tax-free. I cant underscore the importance of tax diversification by contributing to both traditional and Roth accounts you're hedging against future tax uncertainty. If tax rates rise in the future which many experts predict having tax-free income from a Roth account protects you against that.

7. Hyper accumulation involves being intentional about where your money is invested. As you progress in your financial journey you must evolve from a saver to an investor. Hyperaccumulation occurs when you are investing 25% or more of your income for retirement. The most successful investors aren't just lucky they are they well informed and disciplined. A critical point is diversification. A well-balanced portfolio reduces risk while maximizing potential growth. By spreading investments across various asset classes, tax strategies, and investment accounts you protect yourself from volatility and future uncertainty. Therefore opening a brokerage account and optimizing your portfolio are crucial. Markets fluctuate and volatility can make it tempting to pull out during downturns but intelligent investors have a long-term mindset. Three bucket for tax efficiency https://youtu.be/f4_Xr_IPkms?si=Br8TQAlGMw1cPkqw&t=150s

   Because the Hyper accumulation stage is about being intentional about your money, it warms of falling into the Trap of Lifestyle inflation where increase in income leads to higher expenses rather than greater savings. Hyper accumulation thrives on maintaining a balance between enjoying your present life and securing your future by avoiding unnecessary spending and channeling surplus income into Investments you keep your financial growth on track. Beyond retirement accounts the chapter highlights opportunities such as taxable brokerage accounts, real estate or even alternative Investments like private Equity.

8. This extends steps 1 and 4. By strategically planning and putting money aside for expenses you know are coming you reduce your reliance on debt and position yourself for long-term success. Prepaying future expenses it's about removing Financial burdens before they even exist. You know that you will need a car soon. You know that your car will need to have a major repair soon. You know that you home will need some major update or repair soon. This proactive approach to finances doesn't just focus on saving money for the sake of accumulating wealth but on strategically investing in the future you want to create. Like you want your kids to go a private school or to go to college, you want to buy your dream home, travel the world, go on missionary trips, or fund a wedding. Instead of waiting for your bills to pile up or relying on debt to manage future costs you actively set money aside to fund those obligations before they arrive. This could mean saving in a 529 plan, or if they have earned income, custodial Roth IRA.

9. The Silent Drain of Low-Interest Debt

   Low-interest debt—such as student loans, mortgages, or car loans—may seem less pressing than high-interest credit card debt, but it can still be a significant drain on your wealth. Even though the interest rates may appear manageable, the reality is that this type of debt prevents you from fully harnessing the power of your money.

   Every month, a portion of your income is committed to debt payments, which reduces the amount you can save or invest. Over time, this "slow leak" in your finances can add up, preventing you from reaching your financial goals as quickly as possible.

   The chapter points out that while it may feel okay to have low-interest debt, it is still holding you back. The interest paid is money that could have been put to better use, whether for investing, saving, or funding your "abundant goals." The longer you drag out paying off low-interest debt, the more you limit your ability to grow wealth.

WHERE TO INVEST?

- Account Types: Taxable, Tax Deferred, Tax-Free Accounts
  - IRA, 401k, 529, Cash Mgmt. Acc., Brokerage Account, …
- Brokerage Firms: Vanguard, Fidelity, Charles Schwab, eTrade, RobinHood, …
  - Assests Under Management, Lowest Cost and Tax Efficiency
- Management Types: Activate, Passive (Index) Products
  - Passive is cheaper and tends to out perform activate management
  - Investment products: Mutual Funds, ETF, Stocks, Bonds, …
    - ETFs: VOO, SPY, FXIAX, …
    - Stocks: ARM, NVDA, AAPL, …
    - Bonds: Junk bonds, TIPS, I Bonds, …

Add links to rob Bergers vids on portfolio complexity, hidden cost of advisors

CASH MANAGEMENT ACCOUNT

	CD	High Yield	Cash Mgmt
Liquidity	Limited	High	High
Minimum Req	$500-$1000	$5	$0
Inflation Risk	Not inflation protection	Promotional Rates	Moves with Inflation
Opportunity Cost	Locked	Not Locked	Not Locked
Flexibility	Less Flexible	High Flexibility	High Flexibility
Taxes	Ordinary Tax	Ordinary Tax	Qualified Dividends

DETAILED COMPARISON (RECOMMEND READ) CURRENTLY MUTUAL FUNDS IN A CASH MANAGEMENT ACCOUNT ARE THE BEST PLACE TO HAVE CASH BUT THAT IS NOT ALWAYS THE CASE.

CMA	INCOME	TAX	INVEST	TAX	INVEST	TAX	INVEST	RETRIEVE	TAX

https://investor.vanguard.com/accounts-plans/vanguard-cash-plus-account
https://www.fidelity.com/spend-save/fidelity-cash-management-account/overview

CASH MANAGEMENT ACCOUNT

Vanguard Cash Plus

- Next Day Transfers
- Pay bills
- FDIC and SPIC Insured
- Most Tax Efficient Mutual Funds
- Best Rates
- Paypal and Venmo
- Lowest Expense Ratios (0.07-0.12%)
- No Fees*

Comparison of Mutual Funds

Fidelity Cash Management

- Next Day Transfers
- Pay bills
- FDIC and SPIC Insured
- Debit Card
- Free Check-writing
- Good Rates
- Paypal and Venmo
- Highest Expense Ratios (0.42-0.47%)
- No Fees*

Comparison of Three Accounts

Schwab Bank Investor Checking

- Next Day Transfers
- Pay bills
- FDIC Insured
- Debit Card
- Free Check-writing
- Lower Rates
- Zelle, Apple Pay, Google Pay, Paypal and Venmo
- Modest Expense Ratios (0.34%)
- No Fees*
- No Overdraft Fees

AMERICANS HAVE AN ESTIMATED $1.7 TRILLION SITTING IN SAVINGS ACCOUNTS THAT COULD BE EARNING WAY MORE INTEREST.

401K, TSP AND 403B

- Benefits
  - Contribution Matching
  - High Contribution Limits
  - Tax Deferred (Tax Reduction) Traditional
  - Roth Conversion
- Cons
  - Limited Investment Selection
  - Penalties if retrieved before 59 1/2
  - Traditional has Required Minimum Distribution at 72
  - Some Have a Vetting Period

BROKERAGE ACC.	INCOME	TAX	INVEST	TAX	INVEST	TAX	INVEST	RETRIEVE	TAX
TRADITIONAL 401K	INCOME	INVEST	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	TAX
ROTH 401K	INCOME	TAX	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	NO TAX

https://www.irs.gov/newsroom/401k-limit-increases-to-24500-for-2026-ira-limit-increases-to-7500
https://www.youtube.com/watch?v=sFaYjRyoFXw

HOW MUCH TO INVEST?

- **Evaluate Employer Matching**
- **Careful to not over contribute**

Monthly Contributions

Income	10.000000	12.000000	13.000000	14.000000	15.000000	16.000000	17.000000	18.000000
156,000.00	1,300.00	1,560.00	1,690.00	1,820.00	1,950.00	2,041.67	2,041.67	2,041.67
208,000.00	1,733.33	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
312,000.00	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
416,000.00	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
520,000.00	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67
624,000.00	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67	2,041.67

Annual Contributions

Income	10.000000	12.000000	13.000000	14.000000	15.000000	16.000000	17.000000	18.000000
156,000.00	15,600.00	18,720.00	20,280.00	21,840.00	23,400.00	24,500.00	24,500.00	24,500.00
208,000.00	20,800.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00
312,000.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00
416,000.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00
520,000.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00
624,000.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00	24,500.00

Annual Employer Contributions

Income	1.000000	2.000000	3.000000	4.000000	5.000000	6.000000	7.000000	10.000000
156,000.00	1,560.00	3,120.00	4,680.00	6,240.00	7,800.00	9,360.00	10,920.00	15,600.00
208,000.00	2,080.00	4,160.00	6,240.00	8,320.00	10,400.00	12,480.00	14,560.00	20,800.00
312,000.00	3,120.00	6,240.00	9,360.00	12,480.00	15,600.00	18,720.00	21,840.00	31,200.00
416,000.00	4,160.00	8,320.00	12,480.00	16,640.00	20,800.00	24,960.00	29,120.00	41,600.00
520,000.00	5,200.00	10,400.00	15,600.00	20,800.00	26,000.00	31,200.00	36,400.00	47,500.00
624,000.00	6,240.00	12,480.00	18,720.00	24,960.00	31,200.00	37,440.00	43,680.00	47,500.00

Annual Total Contributions

Income	11.000000	14.000000	16.000000	18.000000	20.000000	22.000000	24.000000	28.000000
156,000.00	17,160.00	21,840.00	24,960.00	28,080.00	31,200.00	33,860.00	35,420.00	40,100.00
208,000.00	22,880.00	28,660.00	30,740.00	32,820.00	34,900.00	36,980.00	39,060.00	45,300.00
312,000.00	27,620.00	30,740.00	33,860.00	36,980.00	40,100.00	43,220.00	46,340.00	55,700.00
416,000.00	28,660.00	32,820.00	36,980.00	41,140.00	45,300.00	49,460.00	53,620.00	66,100.00
520,000.00	29,700.00	34,900.00	40,100.00	45,300.00	50,500.00	55,700.00	60,900.00	72,000.00
624,000.00	30,740.00	36,980.00	43,220.00	49,460.00	55,700.00	61,940.00	68,180.00	72,000.00

HEALTH SAVING ACCOUNT

In order to contribute to an HSA, you have to be enrolled in an HSA-eligible health plan. Health savings accounts (HSAs) let you save and pay for qualified medical expenses with tax-free dollars (pre-tax, no tax on growth and no tax at

Every year, the Internal Revenue Service (IRS) sets the maximum that can be contributed to an HSA. For example, for the year 2026 the contribution limit is $4,400 and your employer contributes $1,000, you can contribute $3,400—unless you're eligible for a catch-up contribution of $1,000.

In 2026, you can contribute up to $4,400 if you have health coverage just for yourself or $8,750 if you have coverage for your family.

At age 55, individuals can contribute an additional $1,000

HSA can be coupled with LP-FSA to get $3,400 extra tax free money

HSA Tax Reduction		
2026 Single		Tax Saving
$4,400.00	0.10%	$440.00
$4,400.00	0.12%	$528.00
$4,400.00	0.22%	$968.00
$4,400.00	0.24%	$1056.00
$4,400.00	0.32%	$1408.00
$4,400.00	0.35%	$1540.00
$4,400.00	0.37%	$1628.00
2026 Married		Tax Saving
$8,750.00	0.10%	$875.00
$8,750.00	0.12%	$1050.00
$8,750.00	0.22%	$1925.00
$8,750.00	0.24%	$2100.00
$8,750.00	0.32%	$2800.00
$8,750.00	0.35%	$3062.50
$8,750.00	0.37%	$3237.50

https://www.fidelity.com/learning-center/smart-money/hsa-contribution-limits

HEALTH SAVING ACCOUNT

HSA	INCOME	INVEST	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	NO TAX
TRADITIONAL	INCOME	INVEST	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	TAX
ROTH	INCOME	TAX	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	NO TAX

When you turn 65, you can use HSA money for ineligible expenses penalty free, though you'll have to pay income taxes. That is, you can do what ever you want with the money.

If you exceed the annual maximum contribution limit, you may face a 6% excess tax on your excess contributions in the year you over-contributed and in each year you fail to remove the excess contribution and its earnings. The excess contribution is also considered taxable income. If you correct the error before the tax filing deadline for the year, you may be able to avoid income tax and the excise tax for that year.

529 SAVINGS ACCOUNT

529 plans are tax-advantaged accounts designed specifically for education savings. 529 funds can be used for a wide range of education expenses, including college expenses at postsecondary schools nationwide, tuition for K-12 schools, certain apprenticeship costs, and student loan repayments.

The 529 account beneficiary can be changed to an eligible family member to use for their qualified education expenses.

Earnings grow tax deferred and withdrawals for qualified education expenses are are free from federal income tax and some state taxes.

You can gift money to an existing 529 plan account. That is, you can get a tax write-off for gifting money to someones 529 account.

Withdrawals from a 529 plan account can be taken at any time, for any reason. But, if the money is not used for qualified education expenses, federal income taxes may be due on any earnings withdrawn. A 10% federal penalty tax and possibly state or local tax can also be added. There are exceptions to the 10% penalty—for instance, if the beneficiary receives a scholarship or attends a US military academy. Any earnings would still be subject to federal income tax and any state and local taxes.

529 can be rolled-over to a ROTH IRA since 1/1/2024

529	INCOME	TAX	INVEST	INVEST	INVEST	INVEST	INVEST	RETRIEVE	NO TAX

https://www.fidelity.com/529-plans/overview,
https://www.fidelity.com/529-plans/what-is-a-529-plan https://investor.vanguard.com/accounts-plans/529-plans

ROTH INDIVIDUAL RETIREMENT ARRANGEMENT (IRA)

- Benefits
  - Unlimited Investment Selection
  - Tax Free Capital Gains
  - No Required Minimum Distribution
  - Investment can be retrieved before 59 1/2 without penalties
  - Back Door ROTH IRA
- Cons
  - Low Contribution Limits
  - Limited by Income Level

Roth IRA Contribution Limits (Tax Year 2025)

Single Filers (MAGI)	Married Filing Jointly (MAGI)	Married Filing Separately (MAGI)	Maximum Contribution for individuals under age 50	Maximum Contribution for individuals age 50 and older
under $138,000	under $218,000	$0	$6,500	$7,500
$139,500	$219,000	$1,000	$5,850	$6,750
$141,000	$220,000	$2,000	$5,200	$6,000
$142,500	$221,000	$3,000	$4,550	$5,250
$144,000	$222,000	$4,000	$3,900	$4,500
$145,500	$223,000	$5,000	$3,250	$3,750
$147,000	$224,000	$6,000	$2,600	$3,000
$148,500	$225,000	$7,000	$1,950	$2,250
$150,000	$226,000	$8,000	$1,300	$1,500
$151,500	$227,000	$9,000	$650	$750
$153,000 & over	$228,000 & over	$10,000 & over	$0	$0

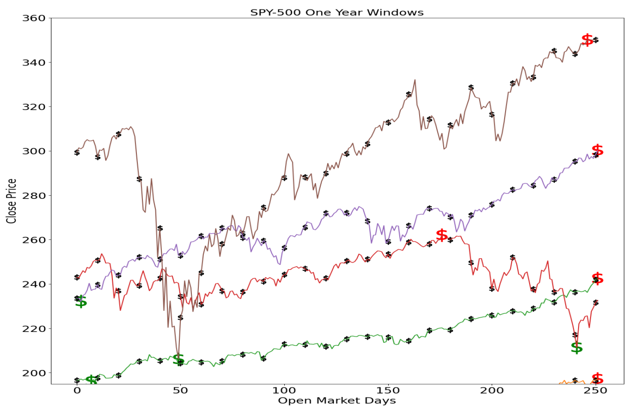
IRA | INCOME | TAX | INVEST | INVEST | INVEST | INVEST | INVEST | RETRIEVE | NO TAX

https://www.schwab.com/ira/roth-ira/contribution-limits

A backdoor Roth IRA is a strategy that allows high-income earners to contribute to a Roth IRA by first making a nondeductible contribution to a traditional IRA and then converting it to a Roth IRA. This method bypasses the income limits that typically restrict direct contributions to a Roth IRA.

WHEN TO INVEST?

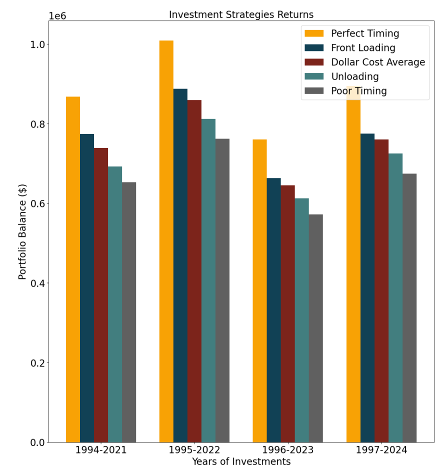
- **Front-loading investments means making a large initial investment upfront**. Front-loading can potentially yield higher returns due to earlier market exposure. This strategy is often used in retirement accounts, such as 401(k)s, IRAs, HSAs, where investors contribute the maximum amount early in the year.
  - **Front-loading Drawbacks:**
    - Risk of missing out on employer matching contributions if contributions are completed early.
    - Less flexibility in adjusting contributions based on market conditions.
- **Dollar Cost Averaging on the other hand, involves spreading out investments over time, contributing smaller amounts regularly rather than a lump sum.** Reduces the risk of investing a large amount just before a market downturn. Allows for adjustments based on market performance and personal financial situations.

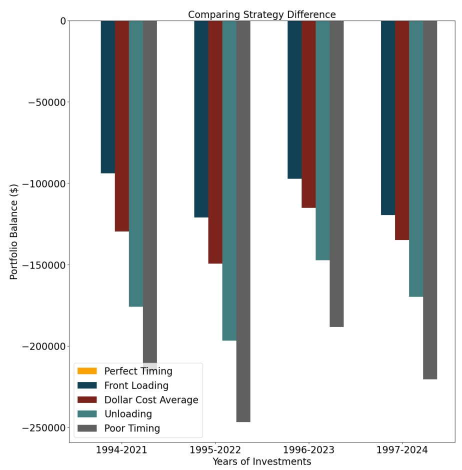


- **Dollar Cost Averaging Drawbacks:**
  - Potentially less growth compared to front-loading if the market rises significantly after the initial investment period.
  - May result in higher overall fees if investing in funds with front-end loads.

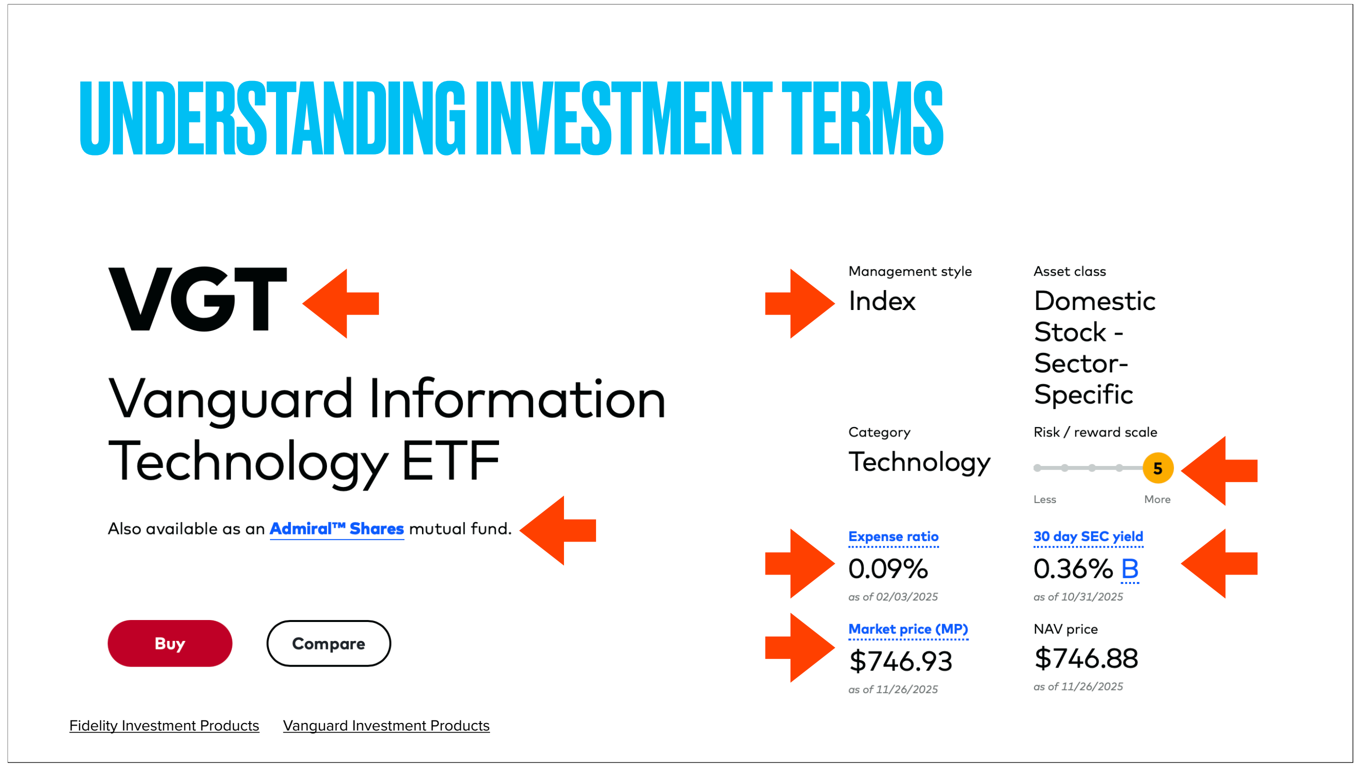
TIMING THE MARKET VERSUS TIME IN THE MARKET

- Investing $7000 for 28 years in the S&P 500. Shifted the start and end year to get more samples.
- Front Loading is best option followed by DCA.
- Poor timing results in large losses
- **73% of the time the market are up. 27% of the time the market are down.**





In a sense, regular investing from your income is a form of unavoidable dollar cost averaging and it serves to smooth the ride. But the big difference is you'll be doing it for many years or even decades to come. It is recommended to put your money to work as soon as you get it in order to have it working for you as long as possible.



Ticker Symbol- name of the ETF and the fact that their is a mutual fund version of this.
Management Style - Active or passive (Index)
Market Price -
Expense Ratio - Expressed as a percentage, the expense ratios are fees charged to investors to cover a fund or ETF's operating costs. They're deducted from dividend and capital gains distributions, not the principal.
30 Day SEC Yield - Distributions and yields that the underlining stocks pay

EXPENSES

- **How are expenses on ETF calculated**

Management Fees	0.09%
12b-1 Distribution Fee	None
Other Expenses	0.01%
Total Annual Fund Operating Expenses	0.10%

Example

The following example is intended to help you compare the cost of investing in the Fund's ETF Shares with the cost of investing in other funds. It illustrates the hypothetical expenses that you would incur over various periods if you were to invest $10,000 in the Fund's shares. This example assumes that the shares provide a return of 5% each year and that total annual fund operating expenses remain as stated in the preceding table. You would incur these hypothetical expenses whether or not you were to sell your shares at the end of the given period. Although your actual costs may be higher or lower, based on these assumptions your costs would be:

1 Year	3 Years	5 Years	10 Years
$10	$32	$56	$128

	Balance_No Expense	Balance_Add Expense	Diff
0	10500.00	10489.50	10.50
1	21682.50	21649.80	32.70
2	33583.99	33516.11	67.88
3	46242.81	46125.36	117.45
35	1224802.98	1198113.79	26689.19
36	1303989.09	1274689.48	29299.61
37	1387403.70	1355282.47	32121.23
38	1475262.27	1440093.44	35168.83

https://personal1.vanguard.com/pub/Pdf/sp958.pdf?2210204664
https://personal1.vanguard.com/pub/Pdf/sp968.pdf?2210204740

EXPENSES

- The gain assuming a fixed maximum contribution of $25,000 over 30 years varies according to the investment expense in a non-taxable account.
- Column “FV” is the Future Value assuming no expense, “FV (1.0)” is the future value after 1.0% compound expense and “1.0” is the compound expense paid. That is, the expense paid $328,893.03 will result in $810,655.19 less gain over 30 years.
- Due to the compounding nature of interest, bigger percentage fees result in lesser gains

No Expense	Balance 0.1% Expense	Diff
26250.00	26223.75	26.25
54206.25	54124.51	81.74
83959.97	83790.27	169.70
115607.02	115313.41	293.61
1678992.11	1652329.84	26662.27
1802180.22	1772410.65	29769.57
1932116.31	1898957.41	33158.90
2069146.62	2032295.44	36851.18

No Expense	Balance 1.0% Expense	Diff
26250.00	25987.50	262.50
54206.25	53391.32	814.93
83959.97	82273.25	1686.72
115607.02	112697.61	2909.41
1678992.11	1433420.70	245571.41
1802180.22	1528886.94	273293.28
1932116.31	1628706.78	303409.53
2069146.62	1733060.94	336085.68

No Expense	Balance 2.0% Expense	Diff
26250.00	25725.00	525.00
54206.25	52581.90	1624.35
83959.97	80609.31	3350.66
115607.02	109847.06	5759.96
1678992.11	1229268.88	449723.23
1802180.22	1303371.42	498808.80
1932116.31	1380199.73	551916.58
2069146.62	1459841.52	609305.10

EXPENSES

- Table of returns after fees have been applied, starting from no fee to 3 % fee.
- The moral of the history is: “Watch out for those small fees”
- My mom use to say: “Learning does not cost you anything, not learning costs you a lot”

Year	0.0	0.035	0.04	0.08	0.1	0.3	0.8	1.0	1.5	2.0	3.0
1.0	25,300.00	25,291.15	25,289.88	25,279.76	25,274.70	25,224.10	25,097.60	25,047.00	24,920.50	24,794.00	24,541.00
2.0	53,130.00	53,101.67	53,097.62	53,065.25	53,049.07	52,887.37	52,484.10	52,323.18	51,921.86	51,521.93	50,726.25
3.0	83,743.00	83,682.54	83,673.90	83,604.84	83,570.32	83,225.68	82,368.25	82,026.95	81,177.84	80,334.64	78,665.91
4.0	117,417.30	117,309.72	117,294.35	117,171.51	117,110.13	116,497.70	114,977.84	114,374.34	112,876.69	111,394.74	108,477.52
27.0	3,063,828.53	3,043,281.92	3,040,358.74	3,017,081.16	3,005,513.95	2,892,419.68	2,629,197.48	2,531,240.97	2,303,213.65	2,097,358.44	1,743,591.67
28.0	3,395,511.38	3,371,729.59	3,368,346.73	3,341,414.00	3,328,033.98	3,197,340.76	2,894,077.89	2,781,568.42	2,520,452.49	2,285,746.40	1,884,953.31
29.0	3,760,362.52	3,732,895.58	3,728,989.21	3,697,894.72	3,682,451.24	3,531,747.71	3,183,115.40	3,054,175.01	2,755,830.77	2,488,828.62	2,035,786.18
30.0	4,161,698.77	4,130,039.12	4,125,537.26	4,089,709.80	4,071,920.37	3,898,491.81	3,498,513.12	3,351,043.58	3,010,863.14	2,707,751.25	2,196,724.85

https://www.wpcu.coop/helpful-resources/financial-calculators
https://www.calculator.net/financial-calculator.html

TAX EFFICIENCY

- **ROTH, HSA and 529b are more tax efficient than a Traditional, FSA or Brokerage Account**
- **Money Market (specifically Vanguard) are more tax efficient than a CD, Dividend ETF or a High Yields accounts on which you are getting taxes every year for every single dollar.**
- **ETF with low dividend are more efficient than High Dividend ETF, Mutual Fund or Individual stocks**
- **If you are planning to have money in reserve, a VSUXX most prudent place to hold the money at the moment (2025).**
- **If you want to save for children's eduction a 529b with a SP500 ETF is the best location (move it to VSUXX/VTMFX 5 years before needing the money)**

Most Tax Efficient

- Low-yielding cash, money market
- Tax-managed stock funds
- Total-market stock index funds
- Mid-cap and small-cap index funds
- Active stock funds
- Balanced funds
- Taxable bond funds
- REIT funds
- High-yield bond funds

Less Tax Efficient

https://docs.google.com/spreadsheets/d/1CDgZVOXy7QLGTHCp3v00G28lvSmrcuvi_f3hNWGT8Hg/edit?gid=1574271208#gid=1574271208

THE POWER OF TAXES

- Many studies have been done to determine how federal taxes reduce mutual fund returns to fund shareholders. One of the longest studies, commissioned by Charles Schwab, was for the 30-year period from 1963 to 1992. The study found that a high-bracket taxpayer who invested $1.00 in U.S. stocks at the beginning of that period would have $21.89 at the end of the period if invested in a tax-deferred account. Meanwhile, the same $1.00 would have grown to only $9.87 if invested in a taxable account—less than half as much. These figures clearly show that taxes, plus the impact of compounding, make a tremendous difference in after-tax performance
- For maximum tax efficiency in taxable accounts, you should do the following:
- Favor funds with low dividends.
- Favor funds with “qualified” dividends.
- Favor funds with low turnover.
- Favor tax-efficient index funds and tax-managed funds.
- You can control the expenses and the taxes

TYPES OF MARKET RISK

Market risk refers to the potential for losses in investments due to fluctuations in market variables.

1. Equity Risk: The risk of loss due to changes in stock prices or their volatility. You investment losing value when you need it like are retirement age or close to buying a home.

2. Interest Rate Risk: The risk associated with changes in interest rates. Defaulting, reducing the rate or extending the pay period. REITs, High Yield (Junk Bonds) are a common high interest rate risk investments.

3. Currency Risk (Foreign Exchange Risk). The risk of loss due to fluctuations in exchange rates.

4. Commodity Risk: The risk of loss due to changes in commodity (oil, gold, home prices, and agricultural products) prices.

5. Opportunity Risk: While not a traditional market risk, refers to the potential loss of benefits from not pursuing an alternative investment or strategy.

6. Shape Risk: The risk related to changes in the shape of the yield curve. Influences the pricing of interest rate derivatives.

7. Holding Period Risk: The risk associated with the time an investment is held. Longer holding periods can increase exposure to market volatility.

8. Basis Risk: The risk that the price of a hedging instrument does not move in line with the price of the asset being hedged. Affects the effectiveness of hedging strategies.

9. Margining Risk: The risk of uncertain future cash outflows due to margin calls. Impact: Affects leveraged positions in trading.

UNDERSTANDING THESE RISKS HELPS MAKE INFORMED DECISIONS AND MANAGE PORTFOLIOS EFFECTIVELY.

DIVIDEND / HIGH YIELDS ETFS VS EQUITY ETFS

- Reduced Capital Appreciation Potential
  - **Focus on Mature Companies: Dividend ETFs, especially those with high yields, tend to hold stocks of mature, stable companies (often in sectors like utilities, financials, and consumer staples) that generate significant cash flow.** These companies distribute a portion of their profits as dividends rather than reinvesting all of it back into the business for high-speed expansion.
  - **Less Exposure to High-Growth Firms: Pure growth companies (often in technology or biotech) typically reinvest almost all of their earnings back into research, development, and expansion to achieve higher revenue and profit growth.** They pay little to no dividend, which is why they are often excluded from or have a lower weighting in dividend ETFs.
- Sector Concentration Risk
  - **Bias Towards Certain Sectors: To achieve their dividend mandates, these ETFs often have an overweighting in sectors known for stable, consistent cash flow and high dividend payouts (like Utilities and Consumer Staples).**
  - **Limitation on Growth Sectors: Conversely, they often have an underweighting or even exclusion of high-growth sectors (like Technology) that are the traditional drivers of major market rallies.**
  - The Growth "Cost": This concentration means your portfolio may miss out on significant growth driven by innovation and secular trends in the most dynamic sectors of the economy.

COMPARE VGT, SCHD AND VIG IN VANGUARD ETC COMPARE

COMPARE VGT, SCHD AND VIG IN ETF INSIDER

COMPARE VGT, SCHD AND VIG IN ETF RESEARCH CENTER

COMPARE VGT, SCHD AND VIG IN PORFOLIO VISULIAZER

Backtest Portfolio Asset Allocation

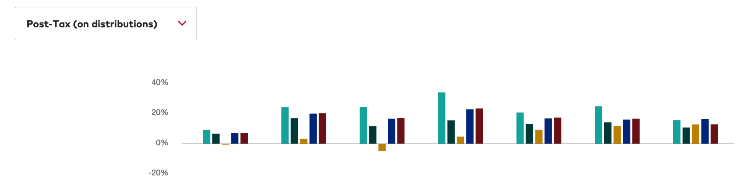
If growth is priority. Dividends and yields are taxed yearly hindering growth. While I can always sell with get the money if I needed it in a low dividend ETF.

The "cost" of investing in dividend ETFs from a growth perspective is primarily the opportunity cost of lower potential capital appreciation compared to pure growth-focused investments.

The Trade-off: By investing in a dividend ETF, you are accepting a lower potential rate of capital gains in exchange for a steadier, often higher, current income stream. During periods when high-growth stocks are outperforming, dividend ETFs will typically lag in terms of total return (capital gains + dividends).

DIVIDEND / HIGH YIELDS ETFS VS EQUITY ETFS

- **Taxable Events: Unlike growth stocks, where you don't pay capital gains tax until you sell the stock, the dividends you receive from a dividend ETF are generally taxed each year.**
- Qualified dividend can be taxed at a capital gains tax rate of 0%, 15%, or 20%.
- Non-Qualified Dividends are taxed as ordinary income (a higher rate). If you don’t hold the shares long enough, the IRS might deem them nonqualified, and you’ll pay the higher, nonqualified tax rate.
- **The Cost: This annual taxation means the power of compounding is slightly reduced, as a portion of your return is diverted to taxes each year instead of being fully reinvested (unless the ETF is held in a tax-advantaged account like a Roth IRA or 401k).**



Fund	3 month	YTD	1 year	3 year	5 year	10 year	Since inception
VGT (as of 11/30/2025)	8.11%	21.38%	21.34%	29.94%	18.16%	21.87%	13.84% (01/26/2004)
VIG (as of 11/30/2025)	5.89%	14.88%	10.33%	13.56%	11.45%	12.48%	9.48% (04/21/2006)
SCHD (as of 11/30/2025)	-0.55%	2.89%	-4.11%	4.15%	8.22%	10.29%	11.24% (10/20/2011)
VOO (as of 11/30/2025)	6.25%	17.50%	14.61%	20.11%	14.83%	14.12%	14.40% (09/07/2010)
S&P 500 (as of 11/30/2025)	6.34%	17.81%	15.00%	20.57%	15.28%	14.63%	11.26% (01/30/1970)

Pre-Tax Returns shown are net of fees and expenses and include the reinvestment of dividends and interest income. The performance shown does not include taxes owed on distributions. **Post-Tax Returns are net of fees and expenses and do adjust for taxes and sales charges.**

Investing in tax inefficient products short term are not the harmful. But long term they are extremely unprofitable. On top of that if you have high cost plus lower returns, you will end up with less money and higher taxes when RMD kick in.

The worst thing is having the tax inefficient products in a taxable account and the tax efficient in a tax free. You are doing it all wrong.

DIVIDEND AND HIGH YIELDS ETFS

- If you are interested in growth. High dividend and high yield ETFs are not the way to go.
- The reason is the tax implication of dividend are almost always taxed at 15%rate while
- The second reason is that at retirement you can to control where you take money from for tax efficiency. If you have 1M and getting 3% which is $30k you are screwed in taxes. While could have taken that money out the 401k that required the minimum distribution.
- It is a bad idea to put a Tax Inefficiency investment in Non-Tax Sheltered account

Tax cost ratio

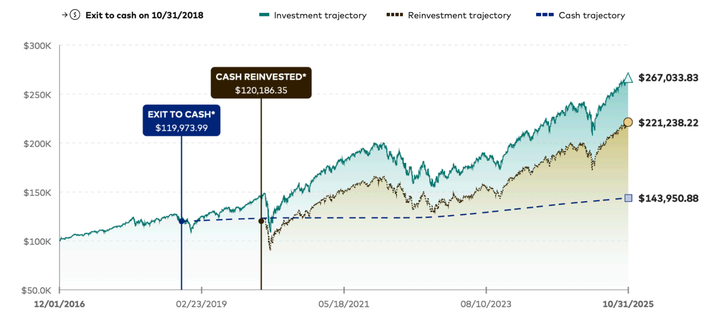
Hide

Fund	1 year	3 year	5 year	10 year
VGT *(as of 11/30/2025)*	0.11%	0.17%	0.20%	0.26%
VIG *(as of 11/30/2025)*	0.42%	0.45%	0.53%	0.54%
SCHD *(as of 11/30/2025)*	1.36%	1.24%	1.15%	1.00%
VOO *(as of 11/30/2025)*	0.31%	0.35%	0.35%	0.41%

The Morningstar Tax Cost Ratio measures how much a fund's annualized return is reduced by the taxes investors pay on distributions. Mutual funds regularly distribute stock dividends, bond dividends and capital gains to their shareholders. Investors then must pay taxes on those distributions during the year they were received. Like an expense ratio, the tax cost ratio is a measure of how one factor can negatively impact performance. Also like an expense ratio, it is usually concentrated in the range of 0-5%. 0% indicates that the fund had no taxable distributions and 5% indicates that the fund was less tax efficient. **For example, if a fund had a 2% tax cost ratio for the three-year time period, it means that on average each year, investors in that fund lost 2% of their assets to taxes. If the fund had a three-year annualized pre-tax return of 10%, an investor in the fund took home about 8% on an after-tax basis. (Because the returns are compounded, the after-tax return is actually 7.8%.**

THE POWER OF EMOTIONS

- **Don't fall for the <u>Cramers</u> yelling to sell. They are pushing you to lock in the lost so they can lock in the wins ("market correction" or "pricing in losses")**
- **"In every market, bulls win, bears win and pigs get slaughtered." - Bloomberg Guest Speaker**
- **Emotions like fear and greed significantly influence stock market behavior, often leading to poor investment decisions. Maintaining a long-term mindset helps investors avoid emotional reactions to market fluctuations, allowing them to focus on their overall financial goals instead.**



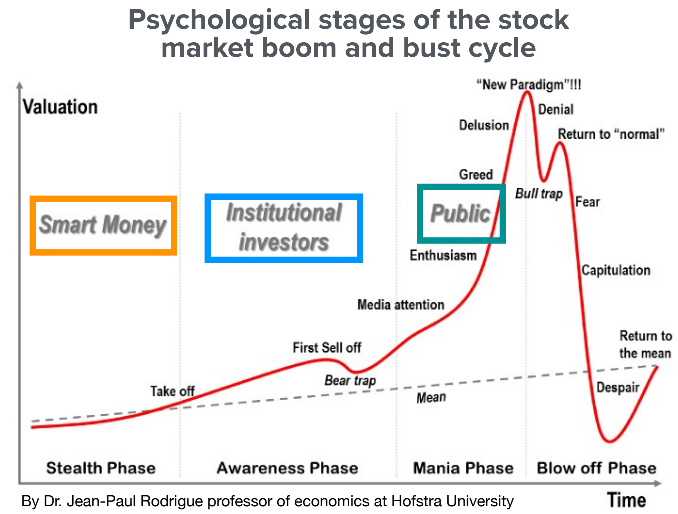
Staying invested resulted in a portfolio ending balance of **$267,033.83, a $123,082.95** gain in portfolio value.

Selling to cash resulted in a portfolio ending balance of **$143,950.88, a $123,082.95** loss in value compared to remaining invested.

Reinvesting cash resulted in a portfolio ending balance of **$221,238.22, a $45,795.61 l**oss in value compared to remaining invested.

THE POWER OF EMOTIONS

- Psychological stages of the stock market boom and bust cycle
- A market bubble is a run-up in stock prices without a corresponding increase in the value of the businesses they represent. In a bubble, speculation and euphoria take over.
- “Everyone is an expert during a bull market”
- These cycle repeat all the time so…
  - Think long term, not short-term
  - Improve your portfolio (Diversification)
  - Buy on sale (Low P/E ratio)



As long as you don’t need to sell assets to raise income to live on, market volatility works for you, not against you… as long as you have the intestinal fortitude to stick with your long-term strategy, and buy the assets you need to support your retirement – now on sale at a reduced price!

This is the time to pick up your fence-sitters, and bring clients over from weak competitors. Nobody switches advisors during bull markets. They change during bear markets! They change because most advisors hide from their phone calls.

Now’s the time to put the pedal to the metal. Meet with or call every client and prospect with a simple message:

Don’t take counsel of your fears.
Think strategy, not short-term tactics.
Diversify.
Pick up great stocks while they’re on sale.
Improve the makeup of your portfolio.
Bear markets are where the best advisors earn their keep. Anybody can look good in a bull market. The worst advisors on the planet can look good during a bull market. Heck, historically, some of them look great during big bull markets! But it’s during the bear markets you can tell who the real pros are.

HOW TO COMPARE INVESTMENT PRODUCTS

- Dont only look at the returns
- Compare expense ratios (lower is better)
- Look at the P/E overtime (lower is better)
- Look at the distribution over time for tax purposes
- 10yr avg return is better than all time avg return statistics due to difference in fund start date
- Compare turn-over ratio (lower is better)
- Compare tax cost ratio (lower is better)

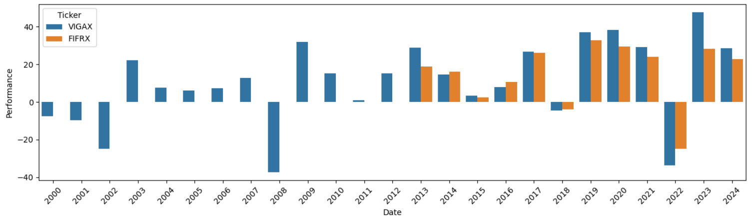
Comparing Averages

	VIGAX	FIFRX
YTD	28.71	22.85
1yr	38.25	25.58
3yr	14.3	8.8
5yr	22.1	15.99
10yr	18.13	14.82
All	10.6	15.27
+ Yrs	19	10
- Yrs	6	2

```
df1 = fun(years,yearly_deposit,13.55,present_value,.47,inflation,age)
df0 = fun(years,yearly_deposit,15.54,present_value,.05,inflation,age)
df0 = df0.merge(df1, on='Age', suffixes=('_VIGAX', '_FIFRX'))
df0.iloc[[0,1,2,3,-4,-3,-2,-1]]
```

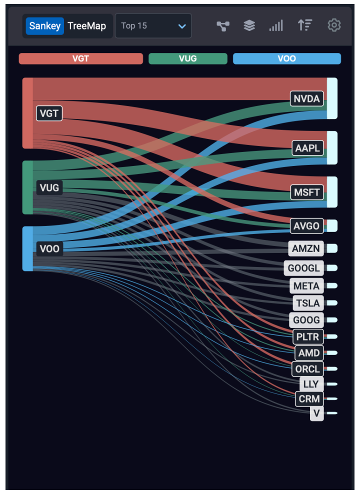
Including Expenses and Inflation

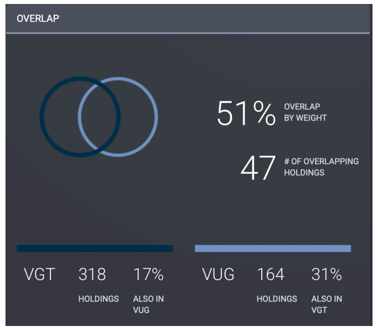
	Age	Balance (Inf. Abj.)_VIGAX	Balance (Inf. Abj.)_FIFRX
0	37.0	343,559.63	336,223.54
1	38.0	468,621.27	450,642.61
2	39.0	605,395.99	572,931.52
3	40.0	755,227.33	703,821.82
27	64.0	16,427,430.88	10,042,497.10
28	65.0	18,286,289.12	10,957,340.59
29	66.0	20,349,290.74	11,950,434.11
30	67.0	22,639,000.52	13,028,587.11



PORTFOLIO OPTIMIZATION AND DIVERSIFICATION

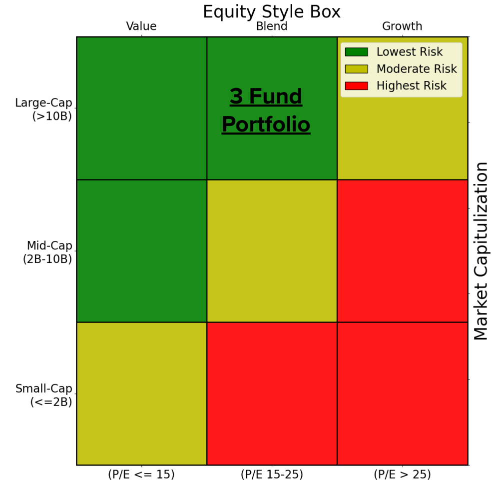
- Find matching funds for tax loss harvesting
- Find mutual excluding funds for diversification
- Customize portfolio for market concentration
- 51% of the total investment between VGT and VUG is in the same companies
- Custom ETF Overlap Calculator



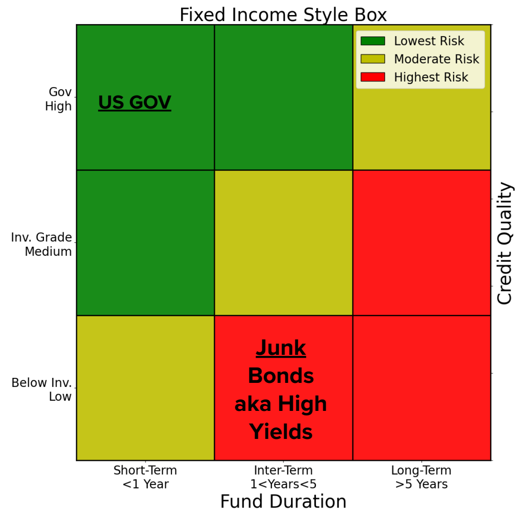


HOW TO COMPARE INVESTMENT PRODUCTS

Stocks



Bonds



https://www.investopedia.com/terms/s/stylebox.asp   Portfolio Visualizer

COMMON MISTAKES

1. Listening to TikTok Financial Influencers
2. Not having the immediate access to accounts (HSA, IRA, 401k)
3. Security is super important (weak password, using android or no multi-factor authentication)
4. Selling in a down turn (Do not sell when markets are readjusting, that is what they want to “price in losses”)
5. You shall not buy the peak on a pump and dump meme stock
6. Not looking at the fundamentals (P/E) and expense but only the stock value
7. Not taking into account the taxes in a taxable account
8. Chasing high yield / junk / high dividend stocks (resulting in higher taxes and less returns with added risk)
9. Not diversifying. High risk investments the first 10 years of investing
10. Timing the market, delayed investment, unnecessary dollar cost averaging ROTH IRA and HSA.
11. Not investing HSA or ROTH IRA (Leaving in settlement account)
12. Not having a cash management account
13. Thinking that past performance is indicative of future performance (This is not basketball)
14. Not taking advantage of all the free information out there including this one :P
15. Not knowing the minimum or maximum contribution limits for the current year
16. The following the investment order as prescribed by Financial Order of Operation (FOO)
17. No awareness of Fidelity, Vanguard, and others investment products (VOOG vs RGAGX vs FSPGX vs Russel 1000 Growth)

THINGS TO WATCH OUT FOR

- Market concentration
- High Expenses
  - Active Managed ETF
- Tax Inefficiencies
  - Dividend and High Yields
- Complex portfolios
  - Dozens of ETFs
- Exceeding Contribution Limits
  - Income Increase, bonus, work change
- Not leveraging taxable and tax sheltered accounts

GOOD RESOURCES

- Useful books to read
  1. The Simple Path to Wealth
  2. Exchange-Traded Funds For Dummies, 3rd Edition
  3. Bogleheads' Guide to Investing
  4. The Little book of Common Sense Investing
  5. Youtube channels: Rob Berger, Ben Felix, Erin Talks Money, The Plain Bagel, Coffeezilla,